

Rean AI Visual Tutor - ERD Package

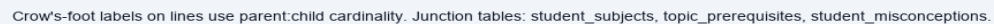
Developer-ready relational ERD with Firestore adaptation notes. This model targets Student and Administrator MVP roles and avoids parent/teacher scope.

Full Logical ERD Diagram

[illegible]

Crow's-foot labels on lines use parent:child cardinality. Junction tables: student, subjects, topic, prerequisites, student_misconceptions.

Simplified MVP ERD - Rean AI Visual Tutor



Crow's-foot labels on lines use parent:child cardinality. Junction tables: student_subjects, topic_prerequisites, student_misconceptions.

Simplified MVP ERD

From	Cardinality	To	Meaning
users	1 to 0..1	student_profiles	user may have one student profile
grade_levels	1 to many	student_profiles	grade has students
student_profiles	1 to many	student_subjects	student selected subjects
subjects	1 to many	student_subjects	subject selected by students
subjects	1 to many	topics	subject topics
grade_levels	1 to many	topics	grade topics
student_profiles	1 to many	tutor_sessions	student sessions
subjects	1 to many	tutor_sessions	session subject
topics	1 to many	tutor_sessions	session topic
tutor_sessions	1 to many	tutor_turns	ordered turns
tutor_sessions	1 to many	student_attempts	session attempts
tutor_turns	1 to many	student_attempts	turn attempts
student_profiles	1 to many	student_attempts	student attempts
tutor_sessions	1 to many	mathematical_verifications	session verifications
tutor_turns	1 to many	mathematical_verifications	step verification
subjects	1 to many	quizzes	subject quizzes
topics	1 to many	quizzes	topic quizzes
grade_levels	1 to many	quizzes	grade quizzes
tutor_sessions	0..1 to many	quizzes	generated follow-up
users	0..1 to many	quizzes	quiz creator
quizzes	1 to many	quiz_questions	quiz questions
quizzes	1 to many	quiz_attempts	quiz attempts
student_profiles	1 to many	quiz_attempts	student quiz attempts
quiz_attempts	1 to many	quiz_answers	attempt answers
quiz_questions	1 to many	quiz_answers	question answers
student_profiles	1 to many	student_topic_progress	topic progress
topics	1 to many	student_topic_progress	topic progress

From	Cardinality	To	Meaning
student_profiles	1 to many	reported_ai_responses	student reports
tutor_sessions	1 to many	reported_ai_responses	reported session
tutor_turns	1 to many	reported_ai_responses	reported turn
users	0..1 to many	reported_ai_responses	assigned admin
tutor_sessions	0..1 to many	ai_request_logs	AI requests

Complete Logical ERD - Entity Inventory

Entity	Primary purpose	Key fields
users	Users	user_id PK firebase_uid UNIQUE full_name email UNIQUE role profile_image_url account_status preferred_language ...
student_profiles	Student Profiles	student_profile_id PK user_id FK UNIQUE grade_level_id FK explanation_level learning_goal onboarding_completed current_streak longest_streak ...
grade_levels	Grade Levels	grade_level_id PK grade_name grade_number UNIQUE description status created_at updated_at
subjects	Subjects	subject_id PK subject_name subject_code UNIQUE icon_url description status display_order created_at ...

Entity	Primary purpose	Key fields
student_subjects	Student Subjects	student_subject_id PK student_profile_id FK subject_id FK selected_at current_progress mastery_level status
curriculum_units	Curriculum Units	unit_id PK subject_id FK grade_level_id FK unit_name description display_order status created_at ...
topics	Topics	topic_id PK unit_id FK subject_id FK grade_level_id FK topic_name topic_code description difficulty_level ...
topic_prerequisites	Topic Prerequisites	prerequisite_id PK topic_id FK prerequisite_topic_id FK relationship_type created_at
lessons	Lessons	lesson_id PK topic_id FK title summary content_reference estimated_duration difficulty_level status ...

Entity	Primary purpose	Key fields
tutor_sessions	Tutor Sessions	tutor_session_id PK student_profile_id FK subject_id FK topic_id FK lesson_id FK NULL original_question detected_language detected_intent ...
tutor_turns	Tutor Turns	tutor_turn_id PK tutor_session_id FK turn_number sender_type message_text stage teaching_strategy interaction_type ...
visualizations	Visualizations	visualization_id PK tutor_turn_id FK UNIQUE visualization_type visualization_data JSON title description render_version validation_status ...
student_attempts	Student Attempts	attempt_id PK tutor_session_id FK tutor_turn_id FK student_profile_id FK submitted_answer answer_format is_correct is_partially_correct ...

Entity	Primary purpose	Key fields
hint_usages	Hint Usages	hint_usage_id PK tutor_session_id FK tutor_turn_id FK student_profile_id FK hint_level hint_type hint_content requested_at
mathematical_verifications	Mathematical Verifications	verification_id PK tutor_session_id FK tutor_turn_id FK verification_type input_expression expected_result generated_result verification_status ...
misconceptions	Misconceptions	misconception_id PK topic_id FK misconception_code misconception_name description recommended_strategy status created_at ...
student_misconceptions	Student Misconceptions	student_misconception_id PK student_profile_id FK misconception_id FK tutor_session_id FK NULL detection_count confidence_score first_detected_at last_detected_at ...

Entity	Primary purpose	Key fields
quizzes	Quizzes	quiz_id PK subject_id FK topic_id FK grade_level_id FK generated_from_session_id FK NULL title description difficulty_level ...
quiz_questions	Quiz Questions	quiz_question_id PK quiz_id FK question_order question_text question_type visualization_data JSON NULL difficulty_level explanation ...
quiz_options	Quiz Options	quiz_option_id PK quiz_question_id FK option_label option_text is_correct display_order
quiz_attempts	Quiz Attempts	quiz_attempt_id PK quiz_id FK student_profile_id FK score correct_count incorrect_count skipped_count started_at ...

Entity	Primary purpose	Key fields
quiz_answers	Quiz Answers	quiz_answer_id PK quiz_attempt_id FK quiz_question_id FK selected_option_id FK NULL submitted_answer is_correct is_partially_correct score_awarded ...
student_topic_progress	Student Topic Progress	progress_id PK student_profile_id FK topic_id FK mastery_score lessons_completed quizzes_completed average_quiz_score correct_attempts ...
learning_activities	Learning Activities	activity_id PK student_profile_id FK activity_type subject_id FK NULL topic_id FK NULL tutor_session_id FK NULL quiz_attempt_id FK NULL title ...
study_streaks	Study Streaks	streak_id PK student_profile_id FK activity_date study_minutes activity_count streak_qualified created_at

Entity	Primary purpose	Key fields
voice_sessions	Voice Sessions	voice_session_id PK student_profile_id FK tutor_session_id FK NULL language session_status started_at ended_at created_at

Entity	Primary purpose	Key fields
voice_interactions	Voice Interactions	voice_interaction_id PK voice_session_id FK audio_input_url transcript_text confidence_score ai_response_text audio_response_url processing_status ...
notification_preferences	Notification Preferences	preference_id PK user_id FK UNIQUE study_reminder_enabled streak_reminder_enabled announcement_enabled reminder_time timezone updated_at
device_tokens	Device Tokens	device_token_id PK user_id FK token UNIQUE platform device_name is_active last_used_at created_at
notifications	Notifications	notification_id PK title message notification_type target_grade_id FK NULL target_subject_id FK NULL scheduled_at sent_at ...

Entity	Primary purpose	Key fields
notification_deliveries	Notification Deliveries	delivery_id PK notification_id FK user_id FK device_token_id FK NULL delivery_status provider_response JSON delivered_at opened_at
reported_ai_responses	Reported Ai Responses	report_id PK student_profile_id FK tutor_session_id FK tutor_turn_id FK report_type description severity verification_status ...
ai_request_logs	Ai Request Logs	ai_request_log_id PK tutor_session_id FK NULL request_type provider model_name prompt_version response_status response_latency_ms ...
ai_error_logs	Ai Error Logs	ai_error_log_id PK ai_request_log_id FK service_name error_type error_message severity retry_status resolved_status ...

Entity	Primary purpose	Key fields
tutor_response_cache	Tutor Response Cache	cache_id PK query_fingerprint UNIQUE normalized_query subject_id FK topic_id FK grade_level_id FK problem_type explanation_level ...
cache_reuse_logs	Cache Reuse Logs	reuse_log_id PK cache_id FK NULL tutor_session_id FK reuse_type compatibility_status rejection_reason latency_saved_ms created_at
audit_logs	Audit Logs	audit_log_id PK admin_user_id FK action_type module target_entity_type target_entity_id previous_data JSON new_data JSON ...
system_settings	System Settings	setting_id PK setting_group setting_key setting_value value_type is_sensitive updated_by FK updated_at

Complete Logical ERD - Relationships

From	Cardinality	To	Meaning
users	1 to 0..1	student_profiles	user may have one student profile
grade_levels	1 to many	student_profiles	grade has students
student_profiles	1 to many	student_subjects	student selected subjects
subjects	1 to many	student_subjects	subject selected by students
subjects	1 to many	curriculum_units	subject units
grade_levels	1 to many	curriculum_units	grade units
curriculum_units	1 to many	topics	unit topics
subjects	1 to many	topics	subject topics
grade_levels	1 to many	topics	grade topics
topics	1 to many	topic_prerequisites	topic prerequisites
topics	1 to many	lessons	topic lessons
users	1 to many	lessons	admin authored lessons
student_profiles	1 to many	tutor_sessions	student sessions
subjects	1 to many	tutor_sessions	session subject
topics	1 to many	tutor_sessions	session topic
lessons	0..1 to many	tutor_sessions	optional lesson grounding
tutor_sessions	1 to many	tutor_turns	ordered turns
tutor_turns	1 to 0..1	visualizations	turn visualization
tutor_sessions	1 to many	student_attempts	session attempts
tutor_turns	1 to many	student_attempts	turn attempts
student_profiles	1 to many	student_attempts	student attempts
misconceptions	0..1 to many	student_attempts	attempt misconception
tutor_sessions	1 to many	hint_usages	session hints
tutor_turns	1 to many	hint_usages	turn hints
student_profiles	1 to many	hint_usages	student hints
tutor_sessions	1 to many	mathematical_verifications	session verifications
tutor_turns	1 to many	mathematical_verifications	step verification

From	Cardinality	To	Meaning
topics	1 to many	misconceptions	topic misconception catalog

From	Cardinality	To	Meaning
student_profiles	1 to many	student_misconceptions	student misconceptions
misconceptions	1 to many	student_misconceptions	detected misconception
tutor_sessions	0..1 to many	student_misconceptions	detection source
subjects	1 to many	quizzes	subject quizzes
topics	1 to many	quizzes	topic quizzes
grade_levels	1 to many	quizzes	grade quizzes
tutor_sessions	0..1 to many	quizzes	generated follow-up
users	0..1 to many	quizzes	quiz creator
quizzes	1 to many	quiz_questions	quiz questions
quiz_questions	1 to many	quiz_options	question options
quizzes	1 to many	quiz_attempts	quiz attempts
student_profiles	1 to many	quiz_attempts	student quiz attempts
quiz_attempts	1 to many	quiz_answers	attempt answers
quiz_questions	1 to many	quiz_answers	question answers
quiz_options	0..1 to many	quiz_answers	selected option
misconceptions	0..1 to many	quiz_answers	answer misconception
student_profiles	1 to many	student_topic_progress	topic progress
topics	1 to many	student_topic_progress	topic progress
student_profiles	1 to many	learning_activities	activity stream
student_profiles	1 to many	study_streaks	daily streaks
student_profiles	1 to many	voice_sessions	voice sessions
tutor_sessions	0..1 to many	voice_sessions	linked voice mode
voice_sessions	1 to many	voice_interactions	voice turns
users	1 to 0..1	notification_preferences	preferences
users	1 to many	device_tokens	devices
users	1 to many	notifications	admin created notifications
notifications	1 to many	notification_deliveries	deliveries
users	1 to many	notification_deliveries	user delivery

From	Cardinality	To	Meaning
device_tokens	0..1 to many	notification_deliveries	device delivery
student_profiles	1 to many	reported_ai_responses	student reports
tutor_sessions	1 to many	reported_ai_responses	reported session
tutor_turns	1 to many	reported_ai_responses	reported turn
users	0..1 to many	reported_ai_responses	assigned admin
tutor_sessions	0..1 to many	ai_request_logs	AI requests
ai_request_logs	1 to many	ai_error_logs	AI errors
subjects	1 to many	tutor_response_cache	cache subject
topics	1 to many	tutor_response_cache	cache topic
grade_levels	1 to many	tutor_response_cache	cache grade
tutor_response_cache	0..1 to many	cache_reuse_logs	reuse logs
tutor_sessions	1 to many	cache_reuse_logs	session cache decisions
users	1 to many	audit_logs	admin actions
users	1 to many	system_settings	setting updater

Firestore Adaptation, Constraints, Retention

- Top-level collections: users, student_profiles, grades, subjects, units, topics, lessons, tutor_sessions, quizzes, quiz_attempts, student_progress, misconceptions, notifications, reported_ai_responses, ai_request_logs, tutor_response_cache, audit_logs, system_settings.
- Recommended subcollections: tutor_sessions/{sessionId}/turns, tutor_sessions/{sessionId}/attempts, tutor_sessions/{sessionId}/hints, tutor_sessions/{sessionId}/verifications, tutor_sessions/{sessionId}/cache_reuse_logs, quizzes/{quizId}/questions, quizzes/{quizId}/questions/{questionId}/options, quiz_attempts/{attemptId}/answers, voice_sessions/{voiceSessionId}/interactions, notifications/{notificationId}/deliveries.
- Embedded fields: compact denormalized display snapshots such as subject name/code, grade name, topic name, quiz totals, latest mastery score, latest misconception summary, and notification target labels.
- Reference fields: store IDs such as student_profile_id, subject_id, topic_id, grade_level_id, tutor_session_id, and quiz_id; avoid relying on Firestore document references for portable service integration.
- Denormalized summaries: keep student_profiles.current_streak, student_profiles.total_learning_time, student_subjects.current_progress, student_topic_progress.mastery_score, quizzes.total_questions, and session verification_status updated by backend transactions or idempotent jobs.

Suggested Firestore Composite Indexes

- student_profiles: (user_id), (grade_level_id, account_status) if status is copied from user.
- student_subjects: (student_profile_id, status), (subject_id, status).
- units: (subject_id, grade_level_id, status, display_order).
- topics: (subject_id, grade_level_id, status, display_order), (unit_id, status, display_order).
- lessons: (topic_id, status, difficulty_level).
- tutor_sessions: (student_profile_id, session_status, updated_at desc), (subject_id, topic_id, created_at desc), (verification_status, created_at desc).